

Medical Affairs / MSL Copilot

A Governed Agentic AI Reference Architecture for Life Sciences

Agent 08 of 8 · draft on-label, evidence-grounded HCP responses — MLR review approves before anything ships

HCLS AI Agent Suite

HCLS AI Agent Suite · Built on AWS · June 2026

FROM WEEKS IN MLR TO DAYS

Medical Affairs / MSL Copilot

A governed Medical Affairs copilot that drafts on-label, evidence-grounded responses for HCP interactions with off-label guardrails enforced technically — while MLR review approves everything before it reaches a healthcare professional.

weeks → months

MLR review is the single biggest content-release bottleneck

[sector-press/estimate]

50–70%

potential reduction in time-to-deliver for medical-legal reviews

[industry-research — McKinsey]

\$3B / \$2.3B

record off-label promotion settlements (GSK / Pfizer)

[gov/peer-reviewed — DOJ]

The Issue & The Cost of Doing Nothing

THE ISSUE

- Field medical teams need rapid, on-label, evidence-grounded responses for HCP interactions.
- Off-label guardrails must be technical, not procedural — a policy SOP is not a control.
- MLR cycles routinely stretch from weeks into months, causing missed launch windows.
- Off-label promotion is among the highest-dollar compliance risks in pharma.

THE COST OF DOING NOTHING

Billions in FCA risk

DOJ recovered \$6.8B in total FY2025 False Claims Act recoveries (healthcare a leading category). Record off-label settlements: GSK \$3B, Pfizer \$2.3B, J&J \$2.2B, AstraZeneca \$520M.

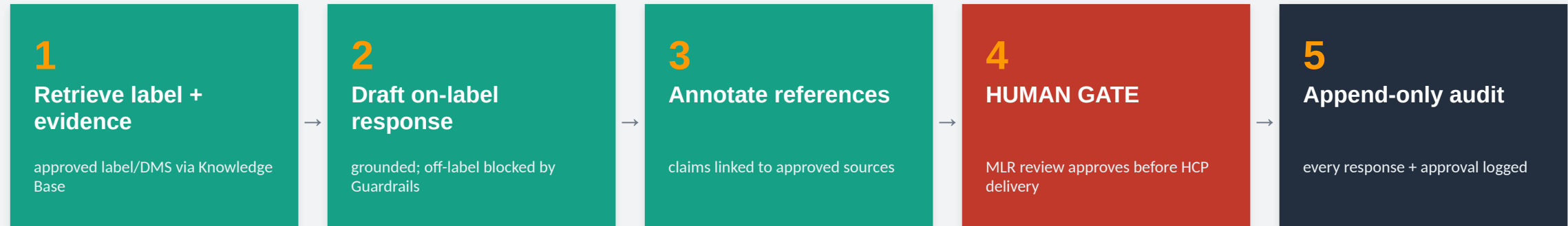
- Off-label promotion liability and corporate-integrity-agreement exposure.
- Missed launch windows when MLR is the bottleneck.

[gov/peer-reviewed — DOJ; industry-research — McKinsey]

The design line: the agent drafts on-label, grounded responses within Guardrails; MLR review approves before anything reaches an HCP — off-label is blocked technically.

How We Solve It — A Governed Pipeline

Retrieve approved label and evidence through a scoped gateway, draft on-label responses within Guardrails — then route every response through MLR before delivery.



Every action audited

node, timestamp, actor, data sources, model + prompt version — append-only and inspection-ready (21 CFR Part 11).

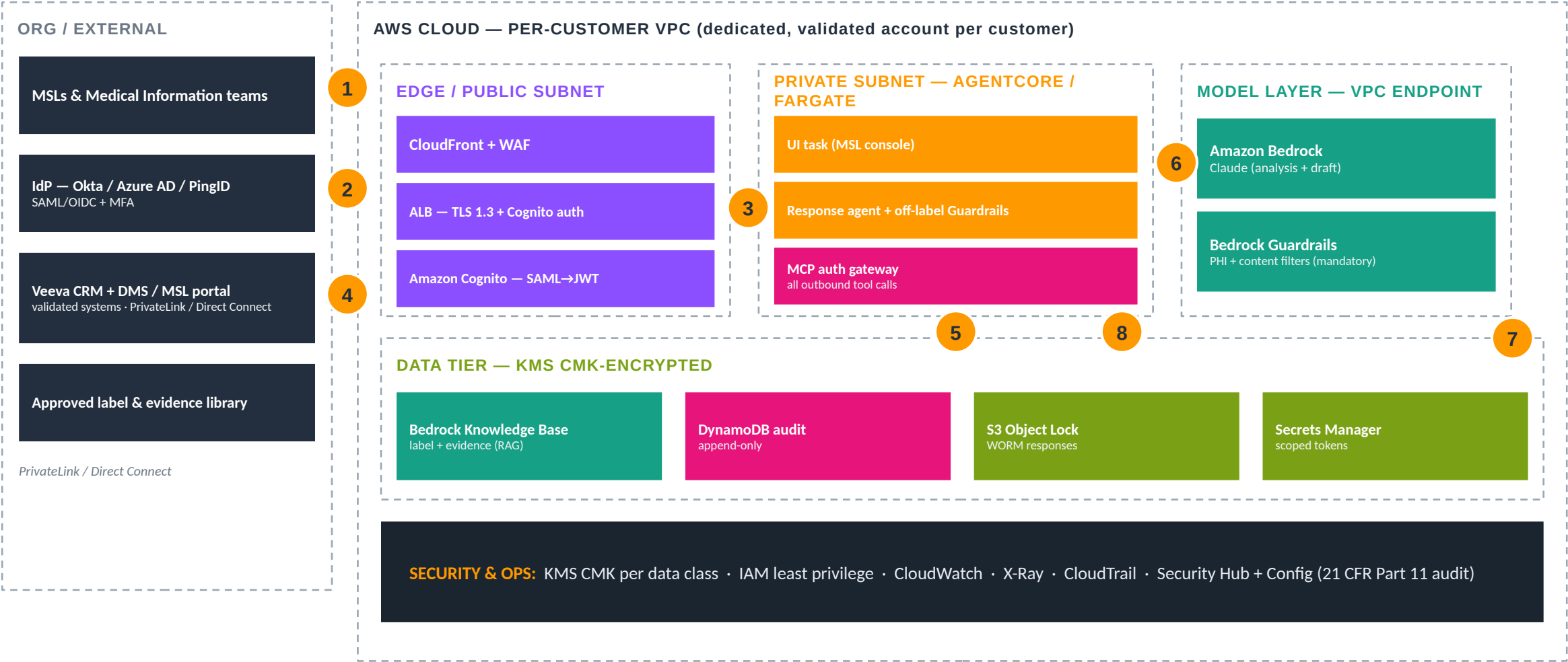
Grounded & explainable

every regulated claim traces to the source corpus; Bedrock Guardrails mask PHI before any model call and bound the output.

AI never decides what reaches an HCP

the consequential action is a human-only step, enforced in the orchestration framework — not a policy SOP.

AWS Architecture & Traffic Flow



1 MSL sign-in 2 SAML federation (MFA at IdP; Cognito issues JWT) 3 authenticated session to MSL console 4 CRM/DMS + label reads over private connectivity 5 demand scales the agent workers 6 model calls stay in-VPC via Bedrock + Guardrails (off-label filters mandatory) 7 every response and MLR approval persisted to append-only audit 8 connectors reachable only through the governed MCP gateway

Proof, Payback & How to Deploy

MEASURED OUTCOMES

50–70%

potential MLR time-to-deliver reduction

[industry-research — McKinsey]

technical

off-label guardrail enforced, not procedural

[design property]

20–30%

medical-writing cost savings (rising as it matures)

[industry-research — McKinsey]

~\$3–5B/yr

potential Medical Affairs efficiency industry-wide

[industry-research — McKinsey]

WHAT IT TAKES TO DEPLOY

1. Provision KMS CMK + per-customer validated VPC / network.
1. Stand up Cognito SAML→JWT federation (MSL / MLR-reviewer roles).
1. Front with CloudFront + WAF, ALB TLS 1.3.
1. Deploy the agent + grant tools via the MCP gateway (deny-by-default).
1. Connect Veeva CRM/DMS; index approved label + evidence; set off-label Guardrails.
1. Enable S3 Object Lock + append-only audit; run smoke + HITL (MLR) test.

Deploy: CloudFormation quick-deploy provisions the isolated environment + dual MCP gateway; connect Veeva CRM/DMS, index the approved label, and set off-label Guardrails.

aws-native-reference/08-medical-affairs-msl/DEPLOY.md · docs/DEPLOY-QUICKSTART.md

The takeaway: the agent isn't the product — the governed platform that makes it 21 CFR Part 11 / GxP-defensible and deployable on AWS is.